theory increasingly becoming a subject on the fringe—the insights of Wess and Zumino launched a tremendous amount of subsequent research on what has come to be called supersymmetric quantum field theory. The supersymmetric standard model, discussed in the preceding section, is one of the crowning theoretical achievements of these pursuits; we now see that, through historical twists and turns, even this point-particle theory owes a great debt to string theory.

With the resurgence of superstring theory in the mid-1980s, supersymmetry has re-emerged in the context of its original discovery. And in this framework, the case for supersymmetry goes well beyond that presented in the preceding section. String theory is the only way we know of to merge general relativity and quantum mechanics. But it's only the supersymmetric version of string theory that avoids the pernicious tachyon problem and that has fermionic vibrational patterns that can account for the matter particles constituting the world around us. Supersymmetry therefore comes hand-in-hand with string theory's proposal for a quantum theory of gravity, as well as with its grand claim of uniting all forces and all of matter. If string theory is right, physicists expect that so is supersymmetry.

Until the mid-1990s, however, one particularly troublesome aspect plagued supersymmetric string theory.

A Super-Embarrassment of Riches

If someone tells you that they have solved the mystery of Amelia Earhart's fate, you might be skeptical at first, but if they have a well-documented, thoroughly pondered explanation, you would probably hear them out and, who knows, you might even be convinced. But what if, in the next breath, they tell you that they actually have a second explanation as well. You listen patiently and are surprised to find the alternate explanation to be as well documented and thought through as the first. And after finishing the second explanation, you are presented with a third, a fourth, and even a fifth explanation—each different from the others and yet equally convincing. No doubt, by the end of the experience you would feel no closer to Amelia Earhart's true fate than you did at the outset. In the arena of fundamental explanations, more is definitely less.

By 1985, string theory—notwithstanding the justified excitement it was engendering—was starting to sound like our overzealous Earhart